

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q21-1739

Roanoke Rapids 345 kV  
August 28, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1739. The Interconnection Customer proposes a 295.6 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 345 kV substation in the North Carolina area. The request was received on May 08, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 542 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,529,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                                    | Contingency                                          | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------------------------------|------------------------------------------------------|-------------|--------------|----------|
| CEDAR FALLS WI – N TOWER TAP 138 KV L                                 | HAGGIN – N TOWER TAP SUBSTATION 345 KV               | 127.6       | 304          | 7.3      |
| WING LANE – SPRING ST SUBSTATION 138 KV line                          | MITCHELL DAM – SKAGGS ISLAND JCT 2 345 KV L          | 109.4       | 309          | 7.5      |
| NORTHERN NECK 345 kV/138 kV Transformer Bank                          | SKAGG ISLAND JCT 2 – CHANCELLOR ROW 345 KV line      | 108.5       | 637          | 5.3      |
| DPC ELROY – N TOWER TAP 138 KV L                                      | MCCARTNEY – N TOWER TAP 345 KV                       | 107.6       | 661          | 6.8      |
| HAGGIN 345/138 kV Transformer Bank                                    | N TOWER TAP – HAGGIN 345 KV                          | 106.9       | 426          | 5.3      |
| REDBED PLAINS WIND FARM – FRUITRIDGE & S. LAND PARK 138 KV            | MITCHELL DAM – SKAGGS ISLAND JCT 2 SUBSTATION 345 KV | 105.8       | 341          | 6.2      |
| IGO – MICKENS 345 KV                                                  | MICKNS – NEWPORT NEWS 345 KV line                    | 105.7       | 1,089        | 7.4      |
| DIST 1000 – BRCHWOOD 138 KV line                                      | BIRCHWOOD – LEAVITT 345 KV                           | 105.4       | 619          | 5.1      |
| FRUITRIDGE & S. LAND PARK – CARLIN SUBSTATION 138 KV                  | MITCHELL DAM – SKAGGS ISLAND JUNCTION 2 345 KV       | 105.4       | 385          | 6.2      |
| TRENTN DIESEL – DIST 1000 SUBSTATION 138 KV L                         | BIRCHWOD – LEAVITT SUBSTATION 345 KV L               | 105.1       | 633          | 5.1      |
| WESTCOAST CELLUFIBRE – REDBED PLAINS WIND FARM SUBSTATION 138 KV line | MITCHELL DAM – SKAGGS ISLAND JCT 2 345 KV line       | 102.0       | 528          | 6.2      |
| MCCARTNEY – GATON 345 KV line                                         | MCCARTNEY – ROANOKE RAPIDS 345 KV                    | 101.6       | 1,659        | 34.9     |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                                              | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| CEDAR FALLS WI – N TOWER TAP 138 KV L                                 | Rebuild line with higher-capacity conductor | \$7,918,000   | 11% (\$870,000)       |
| WING LANE – SPRING ST 138 KV                                          | Reconductor line                            | \$712,000     | 5% (\$32,000)         |
| NORTHERN NECK 345 KV/138 KV Transformer Bank                          | Replace transformer with higher-rated unit  | \$12,017,000  | 6% (\$662,000)        |
| NOTH TOWER TAP – DPC ELROY 138 KV                                     | Reconductor line                            | \$2,050,000   | 5% (\$100,000)        |
| HAGGIN 345 KV/138 KV Autotransformer                                  | Replace transformer with higher-rated unit  | \$12,072,000  | 3% (\$383,000)        |
| REDBED PLAINS WIND FARM – FRUITRIDGE & S. LAND PARK 138 KV            | Reconductor line                            | \$2,773,000   | 3% (\$77,000)         |
| IGO – MICKENS 345 KV                                                  | Reconductor line                            | \$19,120,000  | 4% (\$749,000)        |
| DIST 1000 – BRCHWOOD 138 KV line                                      | Reconductor line                            | \$8,525,000   | 4% (\$383,000)        |
| FRUITRIDGE & S. LAND PARK – CARLIN SUBSTATION 138 KV                  | Reconductor line                            | \$853,000     | 3% (\$24,000)         |
| TRENTN DIESEL – DIST 1000 SUBSTATION 138 KV L                         | Reconductor line                            | \$12,008,000  | 4% (\$540,000)        |
| WESTCOAST CELLUFIBRE – REDBED PLAINS WIND FARM SUBSTATION 138 KV line | Reconductor line                            | \$817,000     | 2% (\$19,000)         |
| GASTON – MCCARTNEY 345 KV                                             | Reconductor line                            | \$37,403,000  | 7% (\$2,690,000)      |
|                                                                       | Total allocated to project                  |               | \$6,529,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.